

Economic Event & Market Data Pipeline

Public revision | Original coursework context and reproducible engineering implementation

Original question

How do global markets behave around economic releases?

Integrate announcements and market observations.

Query structured data and explore event responses.

HSI / EURO STOXX 50: indices; NQ: futures.

Data engineering focus

Explicit CSV contracts and timezone-aware validation.

SQL constraints, transactional upserts, and run records.

SQLite demo + executable MySQL / SQLAlchemy backend.

Streamlit curves, heatmaps, and alignment records.

Original flow

yfinance / investpy -> pandas / CSV -> MySQL -> SQLAlchemy -> Streamlit

Current flow: price connector / event CSV or synthetic generator -> validation -> DB -> reports

investpy collection is historical context; current events enter through a CSV contract.

Relational storage and repeatable ingestion

Historical MySQL design preserved as context; maintained schema is tested on SQLite and MySQL 8.4

Original coursework tables

economic_events: release time, country, importance,
actual / forecast / previous values.
futures_1d_all and futures_1h_all: OHLCV + instrument.
Daily and hourly records were queried from MySQL.

Maintained tables

prices: key = instrument + UTC bar-end timestamp.
events: key = hash(name, country, UTC timestamp).
pipeline_runs: source, timestamp, validated row counts.
Upserts retain unique records on repeated ingestion.

SQL return calculation

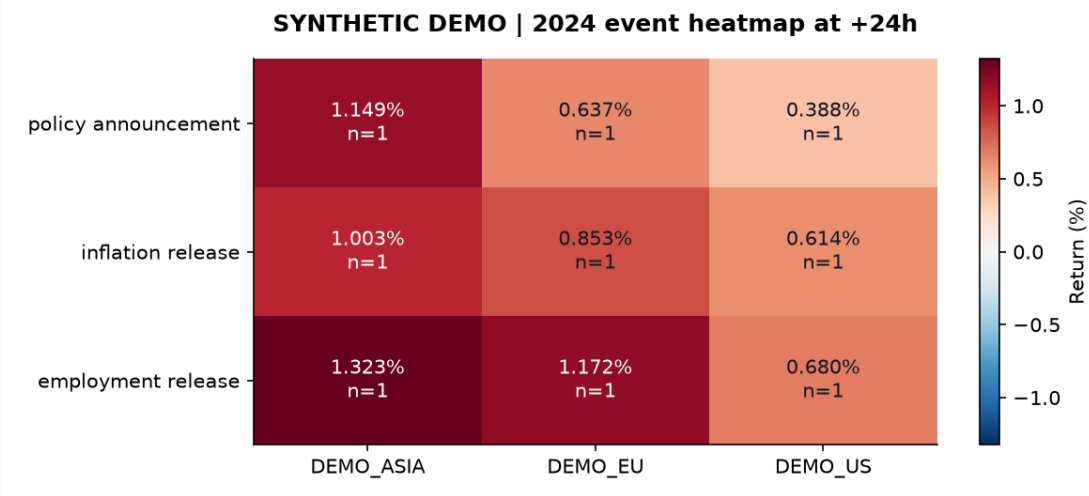
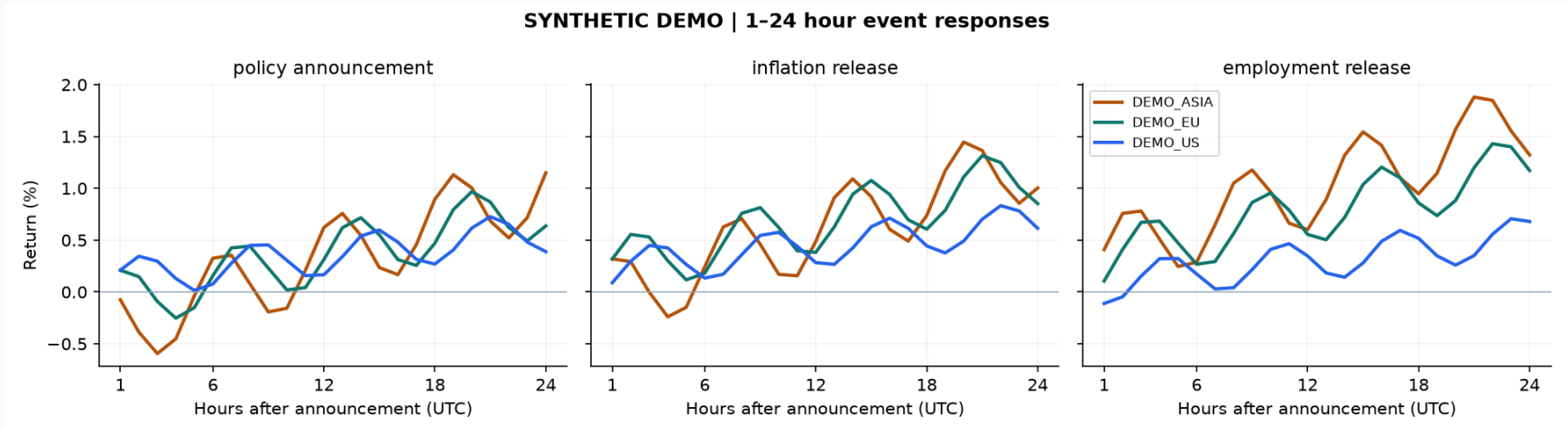
```
LAG(close) OVER (PARTITION BY instrument ORDER BY timestamp_utc)
```

Completed hourly bars | UTC offsets required | Missing/distant observations explicitly excluded

GitHub Actions: duplicate loads, corrections, invalid batches, SQL returns, and event curves.

Event responses and heatmap

SYNTHETIC DEMO ONLY | Generated inputs, not historical market evidence or investment results



288 prices / 3 events / 216 responses

Illustrates the same alignment used by Streamlit.
One synthetic observation per event / instrument.
Heatmap: +24h, not next-day daily close returns.
Tests confirm implementation, not causal impact.
No costs, strategy results, or profitability claims.